Telecom Market in Bangladesh
EDA & Preprocessing — Concept Guide

A line-by-line explanation of every term, library function and logical step used in **EDA_and_Preprocessing.ipynb** and **Assignment.ipynb**. Each concept is given a plain-English definition followed by a worked example taken from the dataset.

Dataset: Telecom Market in Bangladesh — Monthly Subscribers (2008–2015). 6 operators × 96 months = 576 observations across 7 columns after tidying.

Operators covered: Grameenphone, Banglalink, Robi, Airtel, Citycell, Teletalk.

Part 1 — Core Terms & Definitions

1.1 Data & Dataset Structure

Dataset
Definition. A structured collection of related observations stored in rows and columns.
Example. The telecom dataset contains 576 rows where each row is one operator's subscriber count for one month.

Row (Observation / Record)
Definition. A single horizontal entry in a table — one real-world fact.
Example. Row 0 says: *Airtel, Jan-2008, 2.37 million subscribers, market share 6.51%*.

Column (Feature / Variable / Attribute)
Definition. A vertical slice of the table — one measured property repeated across every row.
Example. Subscribers_Mn is the column that holds subscriber count (in millions) for every operator-month combination.

Tidy (Long) Format
Definition. A data layout where each row is one observation and each column is one variable. Easier for grouping, plotting and modelling than 'wide' layouts.
Example. The raw spreadsheet had months as columns (Jan-2008 … Dec-2015). After tidying, every (Operator, Date) becomes its own row.

Wide Format
Definition. A layout where one variable is spread across many columns — typical of spreadsheet exports.
Example. The raw CSV has 96 month columns; tidying via melt reshapes them into a single *Date* column.

1.2 Variable Types

Categorical Variable
Definition. A variable that takes on labels/categories rather than numeric magnitudes.
Example. Operator has six categories: Grameenphone, Banglalink, Robi, Airtel, Citycell, Teletalk.

Nominal Categorical

Definition. Categories with no natural order.

Example. Operator — Robi is not 'greater than' Airtel; they are just names.

Ordinal Categorical

Definition. Categories with a meaningful order but no fixed spacing between them.

Example. Month (January < February < …) and **Year** are ordinal.

Numerical Variable

Definition. A variable measured on a numeric scale where arithmetic makes sense.

Example. Subscribers_Mn, **MoM_Growth_%**, **Market_Share_%**.

Continuous Numerical

Definition. Numerical variable that can take any real value within a range.

Example. Market_Share_% = 16.67 — fractions are legitimate.

Discrete Numerical

Definition. Numerical variable that takes only specific (often integer) values.

Example. Year (2008, 2009, …, 2015) — no values in between.

1.3 Statistical Summary Terms

Count

Definition. The number of non-missing values in a column.

Example. MoM_Growth_% has count = 570 (out of 576) because 6 first rows per operator are NaN — there is no 'previous month' to compute growth from.

Mean (Average)

Definition. Sum of values divided by the count. Sensitive to extreme values.

Example. Mean **Subscribers_Mn** ≈ 14.03 million across all rows.

Median (50th percentile)

Definition. The middle value when data are sorted. Robust to outliers.

Example. Median **Subscribers_Mn** ≈ 8.34 million — much lower than the mean, indicating right-skew (a few giant operators inflate the mean).

Minimum / Maximum

Definition. Smallest and largest values observed.

Example. Min Subscribers = 0.84M (Teletalk early on); Max = 56.54M (Grameenphone, Dec-2015).

Standard Deviation (std)

Definition. Average distance of values from the mean. Larger std = more spread.

Example. Std of **Subscribers_Mn** ≈ 14.11 — comparable to the mean, confirming a wide spread.

Quartiles (Q1, Q2, Q3)

Definition. Values that split sorted data into four equal parts. Q1 = 25th percentile, Q2 = median, Q3 = 75th percentile.

Example. For **MoM_Growth_%**: Q1 = 0.14, Q3 = 2.29.

Interquartile Range (IQR)

Definition. IQR = Q3 - Q1. Captures the middle 50% of the data.

Example. IQR for **MoM_Growth_%** = 2.29 – 0.14 = 2.15 percentage points.

Skewness

Definition. Asymmetry of the distribution. Right-skewed = long tail of high values.

Example. Subscribers_Mn is right-skewed (mean > median) because Grameenphone's huge values pull the tail.

1.4 Data-Quality Terms

Missing Value (NaN)

Definition. An entry where no value is recorded.

Example. MoM_Growth_% is NaN for the very first month of every operator — growth requires a previous month.

Duplicate Record
Definition. Two or more rows that are identical (or share the same key).
Example. df.duplicated().sum() returns 0 — no operator-month appears twice.

Outlier
Definition. A value far from the bulk of the data. May indicate a special event or a measurement error.
Example. 51 rows have **MoM_Growth_%** outside [Q1–1.5·IQR, Q3+1.5·IQR] = [–3.07, 5.50] — including Airtel's –41.7% jump (operator rebranding).

Interpolation
Definition. Estimating missing values from neighbouring observations (commonly linear).
Example. s.interpolate() would fill a missing Robi value in Mar-2010 by linearly averaging Feb-2010 and Apr-2010.

Forward-fill / Back-fill
Definition. Filling NaNs using the last (ffill) or next (bfill) valid value.
Example. .ffill().bfill() handles leading/trailing NaNs that interpolation cannot.

1.5 Engineered Features

Month-over-Month Growth (MoM_Growth_%)
Definition. Percentage change of a value from the previous month: (Vt – V(t–1)) / V(t–1) × 100.
Example. Airtel went from 2.37M (Jan-2008) to 2.60M (Feb-2008): (2.60–2.37)/2.37 × 100 ≈ **9.70%**.

Market Share
Definition. An operator's subscribers expressed as a percentage of total sector subscribers in the same month.
Example. Jan-2008: total sector ≈ 36.43M; Grameenphone ≈ 16.88M → share ≈ 16.88/36.43 ≈ **46.3%**.

Correlation Coefficient
Definition. A value between –1 and +1 measuring linear association between two variables. 0 = no linear relation.
Example. Subscribers_Mn vs **Market_Share_%** = 0.88 (strong positive) — bigger operators have higher share.

Part 2 — Libraries & Key Functions Used

2.1 Libraries

pandas (pd)
Definition. Python library for tabular data. Provides DataFrame (table) and Series (column) objects.
Example. pd.read_csv(...) loads the telecom CSV into a DataFrame.

NumPy (np)
Definition. Numerical computing library. Provides arrays and math utilities.
Example. np.number is used by select_dtypes(include=np.number) to pick all numeric columns.

matplotlib.pyplot (plt)
Definition. Foundational plotting library.
Example. plt.figure(figsize=(11,5)) sets the canvas size before plotting subscriber trends.

seaborn (sns)
Definition. High-level statistical plotting library built on matplotlib.
Example. sns.lineplot(data=df, x='Date', y='Subscribers_Mn', hue='Operator') draws one coloured line per operator.

2.2 pandas Functions in This Project

read_csv
Definition. Loads a CSV file into a DataFrame.
Example. pd.read_csv('telecom_clean.csv', parse_dates=['Date']) — and parse_dates converts the Date string column to real datetime values.

iloc
Definition. Integer-position based indexing — selects rows/columns by number.
Example. raw.iloc[4:10, 2:2+96] selects rows 4–9 and 96 monthly value columns from the raw spreadsheet.

melt

Definition. Unpivots a wide DataFrame into long format.

Example. df.melt(id_vars='Operator', var_name='Period', value_name='Subscribers_Mn') turns 96 month columns into one *Period* column.

to_datetime

Definition. Parses a string column into Python datetime objects.

Example. pd.to_datetime(df['Period'], format='%b-%Y') turns *'Jan-2008'* into *2008-01-01*.

dt accessor

Definition. Convenience methods on a datetime Series (e.g. .dt.year, .dt.month_name()).

Example. df['Date'].dt.month_name() produces 'January', 'February', …

groupby

Definition. Splits the DataFrame into groups based on a key column so an aggregation can be applied within each group.

Example. df.groupby('Operator')['Subscribers_Mn'].mean() returns the average subscribers *per* operator.

transform

Definition. Applies a function per group but returns a Series aligned with the original DataFrame (same length).

Example. df.groupby('Date')['Subscribers_Mn'].transform('sum') gives every row the sector total of its own month — used to compute market share.

pct_change

Definition. Returns the percentage change between consecutive values.

Example. df.groupby('Operator')['Subscribers_Mn'].pct_change() * 100 produces **MoM_Growth_%**.

isnull / sum

Definition. isnull() flags NaNs as True; chaining .sum() counts them per column.

Example. df.isnull().sum() → MoM_Growth_% = 6 missing values.

duplicated
Definition. Marks rows that are duplicates of previous rows.
Example. df.duplicated().sum() = 0 → dataset is unique.

describe
Definition. Generates count, mean, std, min, quartiles and max for numeric columns.
Example. df.describe().round(2) produces the statistical summary table in Task 3.3.

select_dtypes
Definition. Filters columns by their dtype.
Example. df.select_dtypes(include=np.number).columns returns ['Year','Subscribers_Mn','MoM_Growth_%','Market_Share_%'].

quantile
Definition. Returns the value at a given percentile.
Example. df['MoM_Growth_%'].quantile([0.25, 0.75]) returns Q1 and Q3 used in the IQR rule.

corr
Definition. Computes pairwise Pearson correlation between numeric columns.
Example. df[['Subscribers_Mn','MoM_Growth_%','Market_Share_%']].corr() — Subscribers vs Share = 0.88.

agg
Definition. Applies multiple aggregation functions in one call.
Example. df.groupby('Operator')['Subscribers_Mn'].agg(['min','mean','max','std']) compares operators side by side.

to_csv
Definition. Writes a DataFrame to a CSV file.
Example. df.to_csv('telecom_clean.csv', index=False) saves the tidy dataset for re-use in the assignment notebook.

2.3 Plotting Functions

sns.lineplot
Definition. Draws one or more line plots; hue creates a separate line per category.
Example. Used in Task 4.1 to show subscriber growth per operator over time.

sns.boxplot
Definition. Visualises distribution via the median, quartiles, whiskers and dots for outliers.
Example. Task 4.2 — Grameenphone's box is wide (huge growth); Citycell's is narrow.

sns.barplot
Definition. Displays a numeric value per category as bars.
Example. Task 4.3 — average market share per operator.

sns.heatmap
Definition. Renders a 2-D matrix as a colour grid; ideal for correlation matrices.
Example. Task 4.4 — annotated correlation heatmap with the coolwarm palette.

hist
Definition. pandas DataFrame method that plots a histogram for each numeric column.
Example. Task 3.4 — df[[...]].hist(bins=30) shows distribution shape of subscribers, growth and share.

Part 3 — Logic of *EDA_and_Preprocessing.ipynb*

This notebook converts the messy spreadsheet export into a clean, analysis-ready dataset. The logic flows in 10 stages.

Step 1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Why: pandas handles the table, numpy supplies numeric helpers, matplotlib + seaborn are used later for charts. sns.set_style('whitegrid') sets a clean default look; pd.set_option('display.max_columns', 20) lets us see wide tables without truncation.

Step 2. Load Raw Data

```
raw = pd.read_csv('GP Model.xlsx - Telecom Mkt. in Bangladesh.csv', header=None)
```

Why header=None? The source spreadsheet has no clean header row — titles, blank rows and merged cells are scattered across the top. We load everything as raw cells and extract the block we need manually.

Result. A 78×112 grid where rows 4–10 hold operator subscriber counts and columns 2 onward hold months Jan-2008 through Dec-2015.

Step 3. Extract the Subscriber Block

```
years   = list(range(2008, 2016))
months  = ['Jan','Feb',...,'Dec']
date_cols = [f'{m}-{y}' for y in years for m in months]

operators = ['Grameenphone','Banglalink','Robi',
             'Airtel','Citycell','Teletalk']
subs = raw.iloc[4:10, 2:2+96].copy()
subs.columns = date_cols
subs.index   = operators
subs = subs.apply(pd.to_numeric, errors='coerce')
```

Logic. We isolate the 6 operator rows and the 96 month columns, attach meaningful labels, then coerce every value to numeric. errors='coerce' means any stray text becomes NaN (so a bad cell does not crash the pipeline).

Step 4. Reshape to Long (Tidy) Format

```
df = subs.reset_index().rename(columns={'index':'Operator'})
df = df.melt(id_vars='Operator', var_name='Period',
             value_name='Subscribers_Mn')
df['Date']  = pd.to_datetime(df['Period'], format='%b-%Y')
df['Year']  = df['Date'].dt.year
df['Month'] = df['Date'].dt.month_name()
```

Why melt? Statistical and plotting tools work best when each row is one observation. Melting collapses 96 month columns into a single *Period* column, multiplying rows by 96 and producing 6×96 = 576 records.

Why derive Year / Month from Date? So we can group by year (yearly totals) or by calendar month (seasonality) without string parsing.

Step 5. Check Missing Values

```
df.isnull().sum()
```

Logic. Confirms that the raw extraction succeeded. The interpolation guard:

```
df['Subscribers_Mn'] = df.groupby('Operator')['Subscribers_Mn']\
    .transform(lambda s: s.interpolate().ffill().bfill())
```

…is a **defensive** step: *if* any month were missing for an operator, it would be linearly interpolated using only that operator's own time series. groupby('Operator') is critical — interpolating across operators would invent nonsense values.

Step 6. Check Duplicates

```
df.duplicated().sum()
df.duplicated(subset=['Operator','Date']).sum()
```

Both return 0. The second call is a stronger check: it asks whether the same Operator–Date pair appears more than once (which would be a logical contradiction, even if other columns happened to differ).

Step 7. Feature Engineering

```
df['MoM_Growth_%'] = df.groupby('Operator')['Subscribers_Mn']\
                       .pct_change() * 100

sector_total = df.groupby('Date')['Subscribers_Mn']\
                 .transform('sum')
df['Market_Share_%'] = df['Subscribers_Mn'] / sector_total * 100
```

MoM growth uses groupby('Operator') so that Airtel's first month is compared with Airtel's previous month — not with Grameenphone's last month. The result is the percentage change in subscribers from one month to the next.

Market share uses groupby('Date').transform('sum') to attach each row the sector total of *its own month*, then divides. transform (rather than sum) preserves the row count so the result aligns with the original DataFrame.

Step 8. Statistical Summary

```
df.describe()
df.groupby('Operator')['Subscribers_Mn']\
  .agg(['min','mean','max','std']).round(2)
```

Logic. The first call summarises every numeric column across the whole dataset; the second shows the same statistics *per operator* so we can compare market positions directly (e.g. Grameenphone's mean = 36.03M vs Citycell's mean = 1.61M).

Step 9. Outlier Detection (IQR rule)

```
q1 = df['MoM_Growth_%'].quantile(0.25)
q3 = df['MoM_Growth_%'].quantile(0.75)
iqr = q3 - q1
lower, upper = q1 - 1.5*iqr, q3 + 1.5*iqr
outliers = df[(df['MoM_Growth_%'] &lt; lower) |
              (df['MoM_Growth_%'] &gt; upper)]
```

Rule. Any value below Q1–1.5·IQR or above Q3+1.5·IQR is flagged. For MoM growth this gives bounds [–3.07%, 5.50%] and 51 outlier rows — most are early-period high-growth months and operator rebranding events (e.g. Warid → Airtel produced a –41.7% drop).

Important. The notebook *identifies* outliers but does not delete them — telecom growth spikes are real business signals, not data errors.

Step 10. Save Cleaned Dataset

```
df.to_csv('telecom_clean.csv', index=False)
```

The tidy dataset (576 × 7) becomes the input for *Assignment.ipynb*. Saving it ensures the EDA work is reproducible and decoupled from the messy original file.

Part 4 — Logic of *Assignment.ipynb*

This notebook answers the five assignment tasks using the cleaned dataset produced above.

Task 1 — Explain the Dataset

Documents the source (GP / BTRC), the purpose, the row/column structure, and the meaning of every column. This is qualitative context that any reader should know before reading any number.

Task 2 — Identify Categorical vs Numerical

```
num_cols = df.select_dtypes(include=np.number).columns.tolist()
cat_cols = df.select_dtypes(include='object').columns.tolist()
```

Logic. Pandas knows each column's data type. select_dtypes filters them: numeric columns become candidates for arithmetic and correlations, object/string columns become grouping keys.

Note. Although *Year* is stored as int, it is conceptually **ordinal categorical** — we usually group by it, we do not average it.

Task 3 — Exploratory Data Analysis

3.1 Missing values — df.isnull().sum()

Reveals 6 NaNs in MoM_Growth_% (the first observation per operator). This is expected and harmless — pandas drops NaNs from aggregations automatically.

3.2 Duplicates — df.duplicated().sum()

Returns 0. No row is repeated, so counts and means are honest.

3.3 Statistical summary — df.describe().round(2)

Reads at a glance: subscribers span 0.84M → 56.54M, mean 14.03M, std 14.11M. Mean > median for Subscribers_Mn confirms the right-skew driven by Grameenphone.

3.4 Distribution — df[[…]].hist(bins=30)

Histograms show three shapes: **Subscribers_Mn** heavily right-skewed (most observations small, a few very large); **MoM_Growth_%** roughly symmetric and centred near 1% with fat tails; **Market_Share_%** multi-modal — each operator lives near its own share band.

3.5 Correlations — .corr()

Subscribers vs Market_Share = 0.88 (strong, expected). MoM_Growth vs Subscribers ≈ 0.01 — large operators do *not* grow at a systematically different percentage rate.

3.6 Outliers — IQR rule

Identical logic to Step 9 in the preprocessing notebook; confirms 51 outliers and keeps them in the data.

Task 4 — Visualisations

4.1 Line plot — subscribers over time per operator

```
sns.lineplot(data=df, x='Date', y='Subscribers_Mn',
             hue='Operator')
```

hue='Operator' draws one line per operator, creating six time-series in one panel. Reading: Grameenphone leads throughout; Banglalink and Robi rise sharply; Citycell and Teletalk are flat.

4.2 Boxplot — subscriber distribution per operator

Reveals the **spread** of each operator's monthly counts. A tall box = wide range over time (operator grew a lot). Grameenphone's box reaches from ~17M to 56M; Citycell's is tiny.

4.3 Bar chart — average market share per operator

```
share = df.groupby('Operator')['Market_Share_%']\
          .mean().sort_values(ascending=False)
sns.barplot(x=share.index, y=share.values, palette='viridis')
```

Sorting before plotting makes the bars descend left-to-right, which is easier to read. Grameenphone ≈ 42% average share.

4.4 Correlation heatmap

```
sns.heatmap(corr, annot=True, cmap='coolwarm', center=0)
```

annot=True writes the values inside cells; cmap='coolwarm' uses red/blue diverging colours; center=0 anchors zero to white so the sign of the correlation is obvious.

4.5 Yearly sector total — bar chart

```
yearly = df.groupby('Year')['Subscribers_Mn'].sum()
sns.barplot(x=yearly.index, y=yearly.values)
```

Summing within each year aggregates monthly data into one bar per year. The bars almost triple from 2008 to 2015 — the headline market growth story.

Task 5 — Insights

The final markdown cell turns numbers into a story:

- **Leadership.** Grameenphone leads continuously and grew from ~17M to ~56M.
- **Challengers.** Banglalink and Robi rose steadily into clear #2 and #3 positions.
- **Decline.** Citycell shrank; Teletalk grew slowly.
- **Oligopoly.** Top-3 operators control roughly 90% of the market.
- **Growth signal.** Sector subscribers almost tripled in 8 years — consistent with rising mobile penetration in Bangladesh.

Appendix — Quick Reference Cheat Sheet

Symbol / Term	Meaning in one line
NaN	Not-a-Number; pandas' missing-value marker.
DataFrame	A 2-D table (rows × columns) — the central pandas object.
Series	A single column of a DataFrame.
dtype	The data type of a column (int, float, object, datetime, …).
IQR rule	Outlier if value < Q1–1.5·IQR or > Q3+1.5·IQR.
Pearson corr	Linear correlation; values in [-1, 1].
hue=	Seaborn parameter that splits a plot by a categorical column.
groupby	Split-apply-combine on a key column.
transform	Like groupby+apply but returns row-aligned output.
pct_change	Element-wise percentage change vs previous row.
melt	Wide → long reshape.
select_dtypes	Pick columns by dtype family.
describe	Five-number summary + count, mean, std.
MoM growth	(Vt – V(t–1)) / V(t–1) × 100.
Market share	Operator subscribers ÷ sector total × 100.